

AI for All

We Need to Talk, AI: a comic essay, carried into the lecture

Prof. Dr.-Ing. Mark Schutera | September 14, 2026 · 7 parts

Seven parts, one comic

Part	Section	The question it answers
I	Intro	why talk about AI at all
II	Basics	what AI is, and what made it work
III	Examples	where it already runs
IV	Chances	what it is good for
V	Risks	what it costs us
VI	Outlook	who gets to shape it
VII	Social Utopias	what we would want it for

cover, the section spiral

How to read this deck

weneedtotalk.ai



One comic page, one slide, **one quoted passage**. Nothing on the following slides is my wording. The frame title is the page's own heading, the passage is lifted verbatim, and the page number under it tells you where to read the rest.

The comic is a free read at weneedtotalk.ai under CC BY-NC-SA 4.0. It is an essay, not a textbook: it argues. Argue back.

page 5, the authors at the table

[1]

1

Intro

pages 6 to 10

Knowing what future we would like to have

“

Knowing the future is impossible. But what we can do:
Knowing what future we would like to have. And then
work on it.

We Need to Talk, AI | page 6

page 6, the opening spread

[1]

Our creatures have always turned against us

“

In 2011, Stephen Hawking said that ‘Success in creating AI would be the biggest event in history.’ ‘Unfortunately, it might also be the last, unless we learn how to avoid the risks.’

We Need to Talk, AI | page 7

page 7, Golem and Frankenstein

[1]

Who is Julia

“

I love data's friendliness, their lack of strategic answering. If they are biased it's not their fault but ours.

We Need to Talk, AI | page 8

page 8, Julia and the digit block

[1]

Who is Lena

“

AI is a field that gives us an intimate view on society's desires and fears.

We Need to Talk, AI | page 9

page 9, Lena and the visual codes

[1]

Lessons from AI in Fiction

“

Seems, like we should not seek to replicate but to augment ourselves.

We Need to Talk, AI | page 10

page 10, the five fictional robots

[1]

2

Basics

pages 11 to 19

What is Artificial Intelligence?

“

Up to now, it's more adequate to speak of 'artificially augmented intelligence'.

We Need to Talk, AI | page 13

page 13, the essence of a new situation

[1]

Computer Power

“

The amount of computer power used for training AI has gone up to 3,5 month-doubling time.

We Need to Talk, AI | page 14

page 14, the cost and power curve

[1]

Algorithms 1

“

A second key indicator are better algorithms. Now we can teach machines instead of programming them.

We Need to Talk, AI | page 15

page 15, a cat? a cat? a dog..?

[1]

Algorithms 2

“

DL uses artificial Neural Networks (NN) that learn patterns directly from the data they are fed.

We Need to Talk, AI | page 16

page 16, the AI bookshelf

[1]

Data 1

“

Any form of raw fact or figure is data. Whether on paper or in electronic form.

We Need to Talk, AI | page 17

page 17, the data word cloud

[1]

Data 2

“

AI is a journey that begins with data: collecting, storing, transforming and exploring it.

We Need to Talk, AI | page 18

page 18, the data pyramid

[1]

General AI

“

Despite their impressive progress and success, today's AI is narrow. Its tasks are often classification and need a lot of data and a lot of energy.

We Need to Talk, AI | page 19

page 19, the two AI Winters

[1]

3

Examples

pages 20 to 27

Maps

“

Using location data from smartphones... the AI-application Maps can analyze how fast traffic is moving... in real-time.

We Need to Talk, AI | page 21

page 21, two phones and the delta

[1]

Recommendation

“

Also, the platforms use the knowledge of our habits to create and offer new products similar to previous successes... or to let AI autonomously stir those components together that have worked great elsewhere.

We Need to Talk, AI | page 22

page 22, the component tag cloud

[1]

Social Networks

“

Those of us who use social networks use AI frequently,
without necessarily knowing it.

We Need to Talk, AI | page 23

page 23, what a wonderful match

[1]

Self-driving Cars

“

Moreover, user tests show that we need autonomous systems to explain their decision to feel safe. Not easy.

We Need to Talk, AI | page 24

page 24, the unforeseen circumstance

[1]

Identifying Images

“

Unfortunately, AI is susceptible to hallucinations: caused by biased data or by hackers who, for example, make AI mistake guns for apes.

We Need to Talk, AI | page 25

page 25, rifle 2017 and Picasso 1910

[1]

Emotional AI

“

Only $\sim 10\%$ of the emotional meaning of a message is conveyed through words.

We Need to Talk, AI | page 26

page 26, the tone of voice panel

[1]

Exoskeletons

“

Our bodies are different and constantly changing. Nowadays, the only feasible way of translating this to robotics is through the help of AI.

We Need to Talk, AI | page 27

page 27, elderly man or firefighter

[1]

4

Chances

pages 28 to 36

Dealing with Big Data

“

Even with hundreds of us analyzing patterns in big data, the sheer volume simply overwhelms our capacities.

We Need to Talk, AI | page 29

page 29, hundreds of analysts

[1]

Efficiency

“

Humans achieved, on average, an 85% accuracy rate, the AI 95%. Humans took 92 minutes to complete the task, the AI... 26 seconds.

We Need to Talk, AI | page 30

page 30, twenty lawyers against one AI

[1]

Cognitive Biases

“

And this is what makes our distortions have large-scale consequences. Were they random, they would cancel out.

We Need to Talk, AI | page 31

page 31, heuristics and gut feeling

[1]

Creativity

“

We can also get creative directly with the AI, compiling data and coding it, in a narrow area. But the AI has no judgement.

We Need to Talk, AI | page 32

page 32, just a generic style transfer

[1]

Connection

“

Today, they often act like overpaid bookkeepers: taking in and spitting back data, prescribing drugs, adjusting doses, ordering tests.

We Need to Talk, AI | page 33

page 33, the overpaid bookkeeper

[1]

New Opportunities

“

AI now enables us to teach robots how to move, either together with the robot, or, with jackets and gloves containing sensors.

We Need to Talk, AI | page 34

page 34, the sensor jacket and glove

[1]

Inclusion

“

AI reads to us, it narrates the world around us and help us ‘see’. Speech-to-text AI helps us ‘hear’, too.

We Need to Talk, AI | page 35

page 35, a wall missing several bricks

[1]

Comfort

“

It is precisely these institutions that are social places where we can come together and exchange ideas. Where we can decelerate. ‘Waste’ time.

We Need to Talk, AI | page 36

page 36, mall_blueprint.pdf

[1]

5

Risks

pages 37 to 43

Surveillance

“

That means the AI does not only know where we are. The AI can also distinguish what we are doing, whether we are standing, or eating.

We Need to Talk, AI | page 39

page 39, the stick figures behind the wall

[1]

Opinion-Forming and Media

“

AI influences public opinion-forming and the media. Our public sphere has become fragmented.

We Need to Talk, AI | page 40

page 40, the fragmented public sphere

[1]

Future Work

“

We are better in understanding non-trivial, new situations, building relationships, creating context and meaning. Custom-fit solutions.

We Need to Talk, AI | page 41

page 41, the job ladder

[1]

Data Security and Safety

“

Our blind trust in AI providers is misplaced. We all know that the amount of data that we are implicitly or explicitly forced to leave behind is too great.

We Need to Talk, AI | page 42

page 42, the data letter inbox

[1]

Energy Use

“

But conventional hardware was not designed for AI, hence it needs much more computing power, much more energy, than our brains do.

We Need to Talk, AI | page 43

page 43, save energy, save the planet

[1]

6

Outlook

pages 44 to 49

Overall Human-Machine Intelligence

“

Up to date, there is no smartphone that is especially good at helping us get better ideas or interpret information. There is no interface for that either.

We Need to Talk, AI | page 45

page 45, the smartphone as marketplace

[1]

Feminism

“

We need (different!) women to participate in the development and use of AI. Otherwise we'll just reproduce patriarchal structures.

We Need to Talk, AI | page 46

page 46, the first computers were women

[1]

Digital Capitalism

“

Since today's AI is largely incomprehensible to us, we need to regulate and shape AI as a public service and snatch the omnipotence over AI from the few dominant, digital companies.

We Need to Talk, AI | page 47

page 47, Weber's judge and Marx's worker

[1]

ART: Accountability, Responsibility, Transparency

“

Accountability is the concept that AI should be held responsible for the results of its algorithms.

We Need to Talk, AI | page 48

page 48, the three equal pillars

[1]

ART: Accountability, Responsibility, Transparency

“

But let's not fool ourselves. For this ART of AI, we need a new and more ambitious form of governance.

We Need to Talk, AI | page 48

page 48, training, regulation, awareness

[1]

No Fear of Numbers

“

With improved ‘math-esteem’, more of us could and should discuss chances and risks of AI in our lives.

We Need to Talk, AI | page 49

page 49, 1908 against MCMVIII

[1]

7

Social Utopias

pages 50 to 53

AI today is not Skynet

“

For over 300,000 years we've been hunting with spears;
for less than ten years we've been looking for our next
song with the help of AI.

We Need to Talk, AI | page 51

page 50, the section opener

[1]

It depends on the training data

“

If we train AIs on discriminatory data, they will make discriminatory decisions and reduce, not increase, the opportunities, and resources of the underprivileged.

We Need to Talk, AI | page 52

page 52, the application procedure

[1]

Raising the question, not settling it

“

An AI may not be able to resolve all injustices, but it can raise questions about how we want to live.

We Need to Talk, AI | page 52

page 52, the motorway through the city

[1]

Or a creepy surveillance state

“

Or is this vision making us create a creepy surveillance state? Why should the authorities always be well-meaning? How could we make sure they are?

We Need to Talk, AI | page 53

page 53, the same platform, read twice

[1]

What do you think?

“

But the first step has been made: What can AI help us with? Where do we have to be careful? What do you think? Let your voice be heard and your point of view be seen.

We Need to Talk, AI | page 53

page 53, Tbc.

[1]

Further reading, the book's own

Authors	Title
Agrawal, Gans, Goldfarb	Prediction Machines: The Simple Economics of Artificial Intelligence, 2018
Evans	Broad Band: The Untold Story of the Women Who Made the Internet, 2018
Chollet, Allaire	Deep Learning with R, 2018
Lee	AI Superpowers: China, Silicon Valley, and the New World Order, 2018
Bostrom	Superintelligence: Paths, Dangers, Strategies, 2014
Zuboff	The Age of Surveillance Capitalism, 2018
Russell, Norvig	Artificial Intelligence: A Modern Approach, 2010
Daum	Das Kapital sind wir: Zur Kritik der digitalen Ökonomie, 2017

page 55, the reading list

[1]

If you want to build one

the track



This deck is about what AI does to us. For how it is built, work Google's **foundational courses** track, four courses in order:

Course	Length	Answers
Introduction to Machine Learning	20 min	what ML is
Machine Learning Crash Course	12 modules	how to build one
Problem Framing	45 min	whether to at all
Managing ML Projects	90 min	how to keep it alive

developers.google.com/machine-learning/foundational-courses

a laptop on the same table

[2]

References

- 1 Julia Schneider and Lena Kadriye Ziyal. *We Need to Talk, AI: A Comic Essay on Artificial Intelligence*. Berlin: Dr. Julia Schneider, 2019. ISBN: 978-3-748531-28-9. Free read, CC BY-NC-SA 4.0. <https://weneedtotalk.ai>.
- 2 Google Developers. *Foundational Courses: Machine Learning*. Course hub. <https://developers.google.com/machine-learning/foundational-courses>. 2025.
- 3 Google Developers. *Introduction to Machine Learning*. <https://developers.google.com/machine-learning/intro-to-ml>. 2025.
- 4 Google Developers. *Machine Learning Crash Course*. <https://developers.google.com/machine-learning/crash-course>. 2025.
- 5 Google Developers. *Introduction to Machine Learning Problem Framing*. <https://developers.google.com/machine-learning/problem-framing>. 2025.
- 6 Google Developers. *Managing Machine Learning Projects*. <https://developers.google.com/machine-learning/managing-ml-projects>. 2025.

**Let your voice be heard
and your point of view be seen**

Prof. Dr.-Ing. Mark Schutera | DHBW Ravensburg